STUDY NOTES · FOUNDATIONS OF FINANCE

Corporate Finance for Absolute Beginners

From “what is an asset worth?” to the 2008 crisis — explained in plain language, with diagrams, worked examples and practice problems.

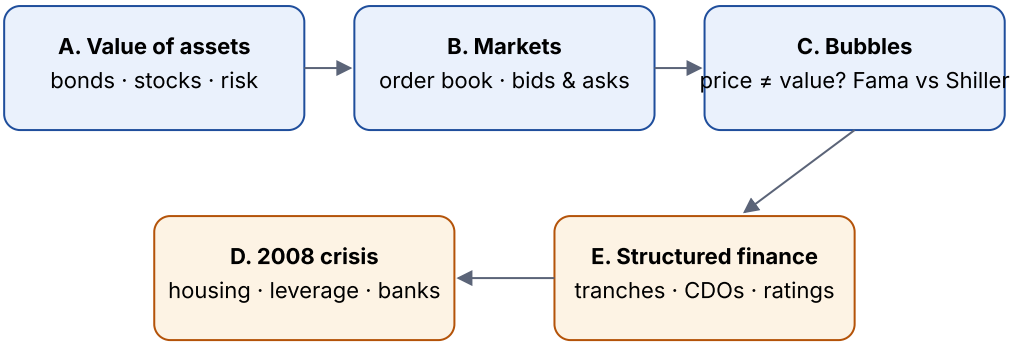
WHAT'S INSIDE

1. **Part A** – How assets get their value (bonds, stocks, risk)
2. **Part B** – How markets trade them (stock exchange, order book)
3. **Part C** – Bubbles & the efficient-market debate
4. **Part D** – The 2008 financial crisis: housing, banks, leverage
5. **Part E** – Structured finance: pooling, tranching, CDOs, ratings
6. **Part F** – Cheat sheet, glossary, 20 practice problems with full solutions

Built from lecture decks FF_1A–1D. Where I added a simple example or formula that is not on the slides, it is marked “extra”.

The Big Picture

Finance answers one question again and again: **"What is something worth today, if what it pays me lies in an uncertain future?"** Everything in these four lectures hangs off that question.



Lessons from A–C (value, risk, bubbles) explain why D and E happened.
E is the "engine"; D is the crash.

Figure 0 – How the four lectures connect

Six words you need before starting

Word	Plain meaning
Asset	Anything you own that can give you money or be sold later (a house, a painting, a bond, a share).
Security	A financial asset that is a tradable piece of paper (or electronic record) — bonds and shares are securities.
Cash flow	Money that an asset pays you, and when (e.g. "Rs. 10,000 in one year").
Interest rate / return	How much extra you earn, as a % of what you paid.
Default	Failing to pay what was promised.
Risk	Uncertainty about whether the cash flows will actually arrive.

PART A · LECTURE 1A

How Assets Get Their Value

1. Why does anyone buy an asset?

You buy a **shop** for two reasons: (1) you can earn rent from it while you own it, and (2) you hope to sell it later, maybe at a higher price. Financial assets work exactly the same way. This second motive — *"I can sell it later"* — is what makes asset markets different from, say, the market for bread (you eat bread; you don't resell it).

KEY IDEA

The value of a financial asset depends on **(1) how big** the expected cash flows are and **(2) how uncertain** those forecasts are. Bigger cash flows → higher value. More uncertainty → lower value.

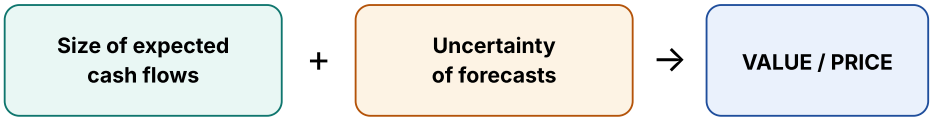


Figure 1 – The two ingredients of value

2. Bonds

DEFINITION

Bond = a security that promises to pay a **fixed amount** of money at **specific dates**. Think of it as an IOU: you lend money now; the issuer repays later.

Take a **one-year bond** that promises Rs. 10,000 one year from today. Suppose today's price is **P**. If you buy it and hold it for a year, your return is the profit divided by what you paid:

$$i = (10{,}000 - P) / P$$

EXAMPLE

Price P = Rs. 9,000. You pay 9,000 and get 10,000 back, so profit = 1,000.
i = 1,000 / 9,000 = **11.1% per year**.
If the price were Rs. 9,500 instead: i = 500/9,500 = 5.3%. **Higher price → lower return.**

Price and interest rate move in opposite directions

Because the bond's payment is *fixed* (always Rs. 10,000), the only thing that can adjust is the price. Rearranging the formula above:

$$P = 10{,}000 / (1 + i)$$

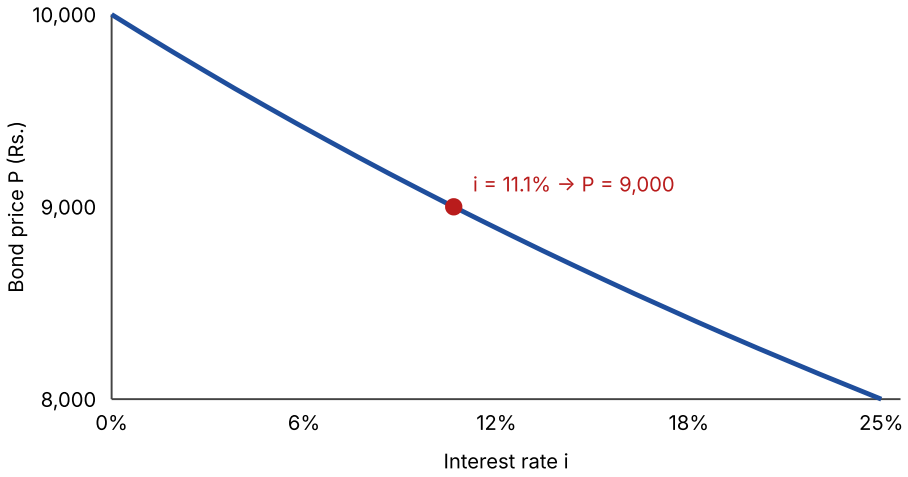


Figure 2 – Bond price falls as the interest rate rises

EXAMPLE – WHY BOND PRICES FALL WHEN RATES RISE

Rates in the economy are 8%. Your bond (pays 10,000 next year) is worth 10,000/1.08 = **Rs. 9,259**. Now rates jump to 10%. Nobody will pay 9,259 for something yielding 8% when they can get 10% elsewhere. The price must drop to 10,000/1.10 = **Rs. 9,091**.

Risky bonds (corporate bonds)

Government bonds: default risk is usually negligible. **Corporate** bonds are not risk-free. Investors demand extra interest to compensate for default risk — and a higher required interest rate means a **lower price**.

RULE

If two bonds promise exactly the same payments, the **riskier one has the lower price**. If it doesn't default you earn more; but you may not get paid at all.

3. Stocks (shares)

DEFINITION

Stock / share = a claim on a part of a firm's assets and hence its profits. Profits are shared in direct proportion to shares held.

A firm's profit is split into two buckets:

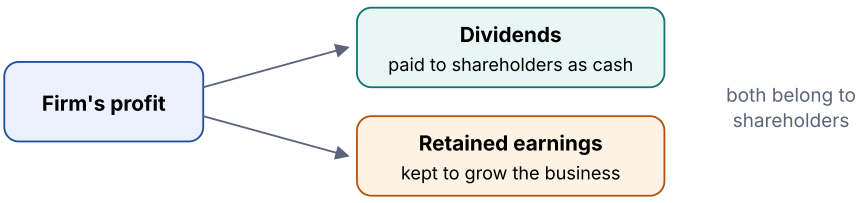


Figure 3 – Where a firm's profit goes

EXAMPLE

A company has 1 crore shares and earns Rs. 10 crore profit, i.e. Rs. 10 per share. You own 1,000 shares → your slice is Rs. 10,000. If the firm pays out 40% as dividends you receive Rs. 4,000 cash; the other Rs. 6,000 stays in the business to (hopefully) raise future profits and the share price.

BOND VS STOCK

	Bond	Stock
What you hold	A loan (IOU)	Part-ownership
Payments	Fixed, promised	Not promised, variable
Timing	Specific dates	Not fixed
Price driven by	Interest rates + default risk	Expected profits + risk

Firms expected to earn more → higher valuation → higher share price.

4. Risk: two very different kinds

Both bonds and stocks derive their value from uncertain earnings. Slide 1A splits that uncertainty in two:

Type	Meaning	Examples	Can you diversify it away?
Systematic (market-wide)	Hits *many* assets at once	Interest-rate change, trade policy, recession, economy-wide demand	**No** – and it is what investors get paid for bearing
Idiosyncratic (firm-specific)	Hits *one* firm/asset	A product succeeds, lawsuit lost, CEO quits	**Yes** – so it's irrelevant for valuation

INTUITION – THE UMBRELLA AND ICE-CREAM SELLER

Own shares in *one* ice-cream firm and one rainy summer ruins you. Own an ice-cream firm *and* an umbrella firm and bad weather for one is good for the other: firm-specific luck cancels out. But if the whole economy crashes, *both* suffer — that's systematic risk, and no amount of mixing removes it.

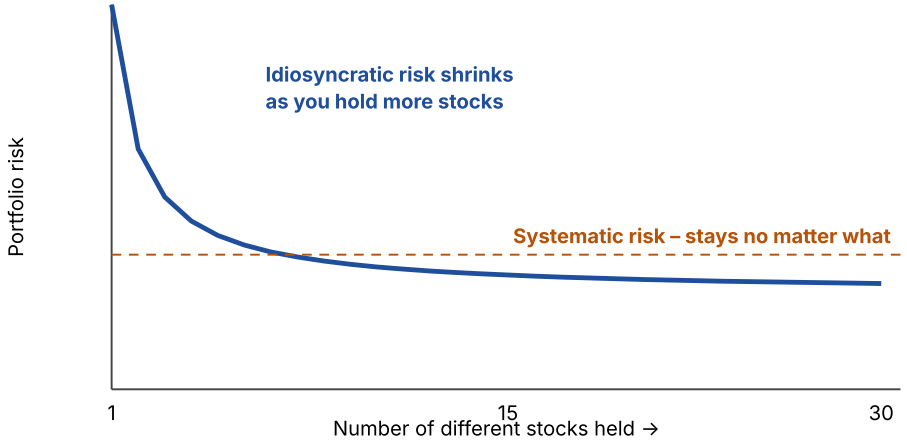


Figure 4 – Diversification (schematic, extra illustration)

Required rate of return (market capitalization rate)

The return that persuades investors to buy a company's shares. It is **higher** for firms with more systematic risk. For the same expected earnings, a **lower** required return means a **higher** share value. (Same logic as bonds: higher required rate → lower price.)

EXAMPLE (EXTRA)

Two firms are each expected to be worth Rs. 1,100 next year. Firm A is very sensitive to the economy; investors demand 20% → price today ≈ 1,100/1.20 = **917**. Firm B is steady; investors demand 10% → price ≈ 1,100/1.10 = **1,000**. Same expected payoff, different price, because of different systematic risk.

5. Trading strategies

- **Fundamental value investing:** estimate what an asset *should* be worth; buy if the price is lower, sell if it is higher. (Used by active mutual funds, some hedge funds.)
- **Momentum trading:** buy because the price is rising (you expect it to keep rising), sell because it is falling. You might even pay more than fundamental value if you think someone will pay you more later.

Both are forms of **speculation**: value investors bet prices *return* to fundamentals; momentum traders bet that short-run trends *persist*. These strategies shape price behaviour — including bubbles and crashes (Part C).

PART B · LECTURE 1A

How Financial Markets Work

1. Primary vs secondary market

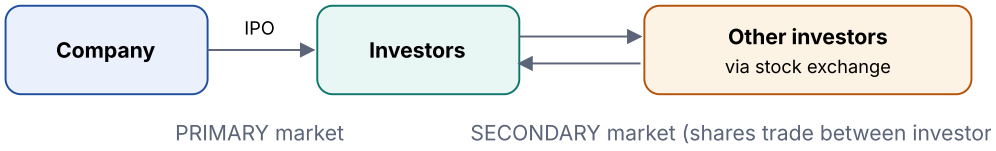


Figure 5 – New shares vs. second-hand shares

- **IPO (initial public offering):** a company sells new shares to the public for the first time.
- Afterwards the shares trade on a **stock exchange** (e.g., Sensex/BSE) — the **secondary market**. The company doesn't get the money from these trades; investors swap shares among themselves.

2. Why prices keep changing: the continuous double auction

RIL's share price moved all through a single day in April 2020. Markets don't run in fixed periods — trading is **continuous**, and buyers and sellers **both** post prices (hence "double" auction).

KEY TERMS

Reservation price: a seller's lowest acceptable price / a buyer's highest acceptable price.
Limit order: a price + quantity you submit. *Buy limit:* "I'll buy up to N shares at no more than Rs. X." *Sell limit:* "I'll sell N shares at no less than Rs. X."
Bid = an order to buy. **Ask** = an order to sell.
Order book: the electronic list of waiting (unfilled) orders. Bids are sorted highest first; asks lowest first.

What happens when you place a limit buy order?

CASE 1 – A MATCHING SELL ORDER EXISTS

If someone is already offering shares at your price *or lower*, a trade happens right away — at the seller's (ask) price, the best deal for you.

CASE 2 – NO MATCH

Your order joins the **order book** and waits until a new sell order arrives that fits.

A tiny order book

ASKS (sellers) – lowest first is "best"	
Price (Rs.)	**Shares**
101.50	300
101.10	200
101.00	100 ← best ask
spread = 101.00 − 100.80 = 0.20	
100.80	150 ← best bid
100.50	200
BIDS (buyers) – highest first is "best"	

Figure 6 – Simplified order book (invented numbers for illustration)

EXAMPLE FROM THE LECTURE (RIL)

A buy order for **1,500 shares at Rs. 1,418.30** matches the cheapest asks first: **50 shares at 1,418.20, 1 share at 1,418.25, and the remaining 1,449 shares at 1,418.30** (50 + 1 + 1,449 = 1,500). You were willing to pay up to 1,418.30 but you got the cheaper ones at the cheaper price.
A buy of 100 shares at 1,418.10, on the other hand, is below every ask, so it **enters the book at the top of the bid column** and waits.

RULE

Immediately-filled buy → trades at the **ask** price (best for buyer). Immediately-filled sell → trades at the **bid** price (best for seller).

How news moves the price

Bad news (rumour a board member will resign): someone submits a big sell order priced below the best bid → it trades against existing bids, using them up, so the price steps **down**. **Good news**: buyers place orders above the lowest ask → they eat through asks, so the price steps **up**.

EQUILIBRIUM?

The price never sits still, but it is always adjusting so that supply and demand match (the market "clears"). The price at any moment reflects everybody — even those just holding.

PART C · LECTURE 1A

Bubbles and Crashes

DEFINITION

Bubble = a sustained and significant departure of an asset's price from its fundamental value. A sudden collapse afterwards is a **crash**.

Case: the tech bubble (Nasdaq, 1995–2004)

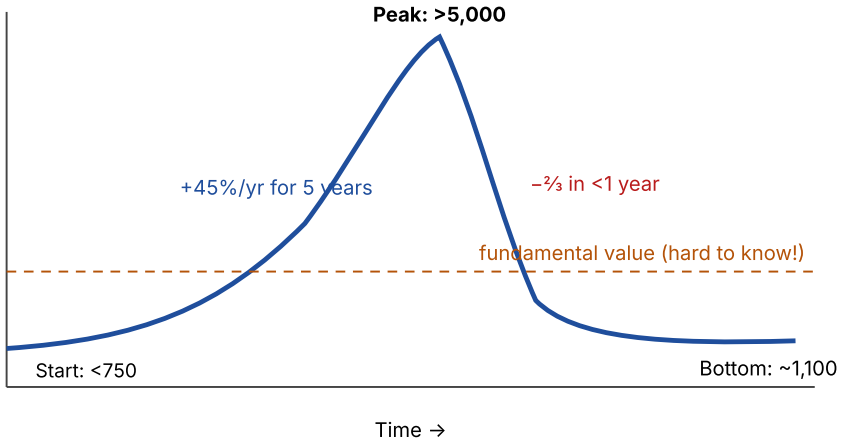


Figure 7 – Shape of the Nasdaq bubble (schematic, numbers from the slides)

CHECK THE MATHS

750 → 5,000 in 5 years: growth factor = 5,000/750 = 6.67. Annual growth g satisfies $(1+g)^5 = 6.67$ → $1+g = 6.67^{1/5} \approx 1.46$, so **≈ 46%/yr** (slides say "around 45%").
Bottom of 1,100 vs peak 5,000: 1 − 1,100/5,000 = 78% down — "almost 80% below its peak".

Three ingredients that breed bubbles

1. **Resale value:** you can speculate on a price change (houses, paintings).
2. **Ease of trading:** in financial markets you can flip from buyer to seller in seconds.
3. **Ease of borrowing:** if you can borrow to buy, you push prices higher and keep the rally going.

The self-reinforcing loop

Prices carry information ("someone knows something"). If traders read a price rise as a sign of *further* rises (momentum), the result is:

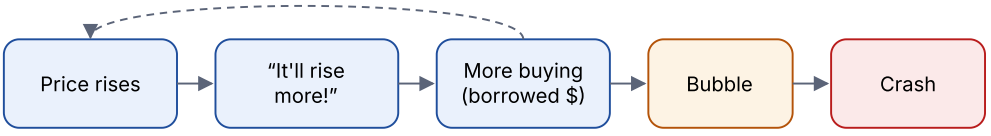


Figure 8 – Momentum loop

Do bubbles even exist? Fama vs Shiller

EFFICIENT MARKET HYPOTHESIS (EMH)

All generally available information about fundamental value is incorporated into prices virtually instantly. If someone had a formula that predicted crashes a week ahead, everyone would use it and prices would drop a week earlier — so such a formula can't reliably exist.

	Eugene Fama (EMH)	Robert Shiller
Bubbles?	"Mostly 20-20 hindsight." People who say prices are too high are right about half the time.	Bubbles can be spotted with simple public stats, e.g. the **price-to-earnings ratio**.
Tech bubble	Prices reflected best available information, which changed sharply.	Prices driven by expectations that they'd keep rising: **"irrational exuberance"**.
Strategy	No reliable way to beat the market.	"Lean against the wind": buy cheap (low P/E), sell dear. Short-run losses, long-run gains beating a diversified basket.

COUNTER-ARGUMENT TO EMH

Even if you *suspect* a bubble, you don't know when others will start selling. Betting against it too early can ruin you, so investors ride it. Citigroup's CEO in 2007: "As long as the music is playing, you've got to get up and dance."

P/E RATIO (EXTRA, SIMPLE)

P/E = share price ÷ earnings per share. A share priced Rs. 500 that earns Rs. 10 per share has P/E = 50: you pay 50 years' worth of current profit. Compare to a typical firm at P/E 15 — Shiller would say 50 looks expensive unless growth is spectacular.

PART D · LECTURE 1B

The 2008 Financial Crisis

Analysts saw US house prices falling from 2006 and expected a slowdown in housing. Few foresaw a **global crisis** — because the key missing piece was how a housing fall would transmit through the **financial system**.

Step 1 – The housing boom and bust

- Early 2000s: interest rates fell a lot → low mortgage rates → more demand for houses → higher prices.
- Financial innovation: **subprime mortgages** — home loans to riskier borrowers.
- While prices rose, it all looked fine (a borrower in trouble could just sell at a profit).
- From 2006 prices fell → many owed **more than the house was worth** → defaults.

EXAMPLE

You buy a house for Rs. 50 lakh with a Rs. 48 lakh loan. Prices fall 10% → house worth 45 lakh but you owe 48 lakh: you are "underwater" by 3 lakh. Walking away (defaulting) can look sensible.

Step 2 – Why banks were fragile: three ingredients

1. HIGH LEVERAGE

Financing assets mostly with borrowed money.

2. ILLIQUID ASSETS

Complex assets hard to sell quickly except at "fire-sale" prices.

3. LIQUID LIABILITIES

Funding via short-term debt that lenders can pull back quickly (wholesale funding). Ordinary checking deposits were insured, but wholesale investors were not — so they ran.

Leverage – the core idea

Leverage = Total assets ÷ Own capital (equity)

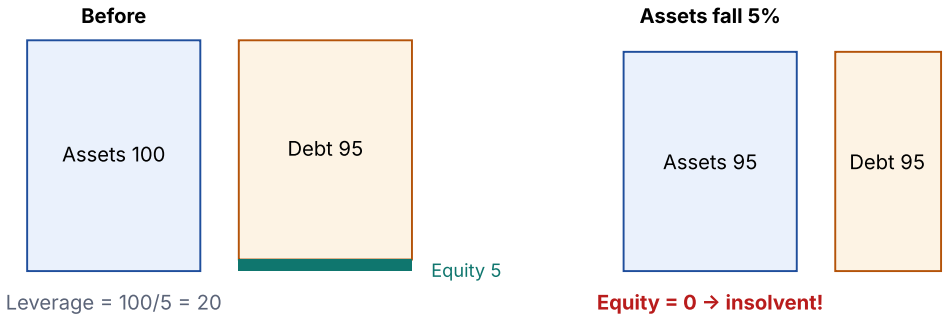


Figure 9 – A 5% drop in asset values wipes out a bank levered 20×

RULE OF THUMB

With leverage L, a fall of 1/L in asset values wipes out your capital entirely. At 20× leverage that's just 5%; at 10× it's 10%.

Why were banks so levered? They **underestimated risk**; managers were **rewarded for chasing high-return projects**; and banks **dodged regulation** using SIVs.

SIVs (Structured Investment Vehicles)

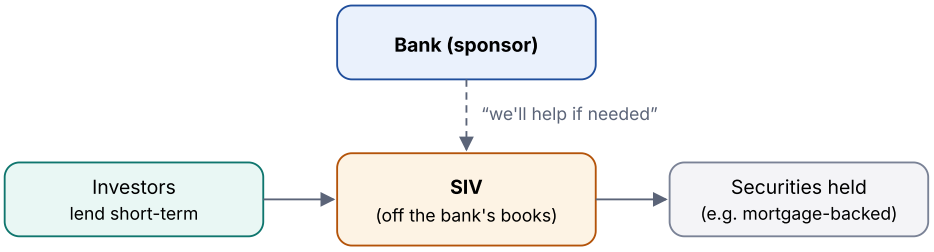


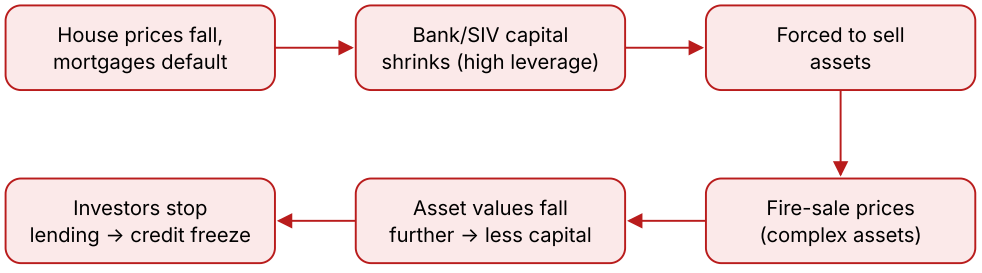
Figure 10 – How a SIV works

A SIV borrows **short-term** from investors (liability side) and holds various **securities** (asset side). The sponsoring bank guaranteed support, yet the SIV stayed off its balance sheet — letting the bank get around leverage rules. When houses fell, people questioned whether SIVs (and the banks behind them) were solvent.

Securitization

Banks pooled mortgages from across the country into **mortgage-backed securities** (MBS). Idea: a local house-price fall won't sink a bank holding loans from everywhere. Problems: banks cared less about whether borrowers could repay (they sold the loans on), and rating agencies mis-valued the securities. (Details in Part E.)

Step 3 – The vicious spiral



...and the loop repeats, each turn making things worse

Figure 11 – The crisis feedback loop

Step 4 – Hitting the real economy

- Interest rates at which people and firms could borrow jumped; confidence collapsed.
- Falling stock and house prices + lower confidence → **less consumption**; also less investment.
- Result: fall in **aggregate demand** (total spending in the economy) → recession.

Step 5 – Policy responses

Tool	What was done
Deposit insurance	Federal deposit insurance expanded (calms depositors).
Liquidity	Federal Reserve lent freely to financial firms so they could borrow.
TARP	Troubled Asset Relief Program – meant to remove complex assets from bank balance sheets; later used to inject capital into banks.
Monetary policy	Fed cut the policy rate to **zero by Dec 2008** – hitting the **zero lower bound** (can't go much below 0), so it had to use **unconventional** policy.
Fiscal policy	Tax cuts and higher government spending to boost demand.

PART E · LECTURES 1C AND 1D

Structured Finance & Credit Ratings

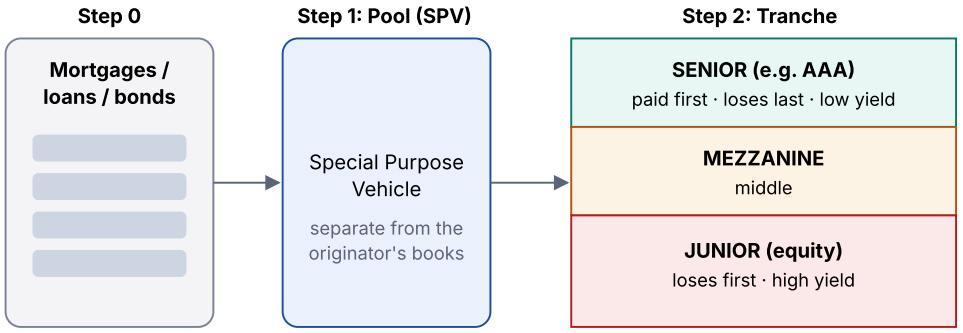
1. Rise and fall in numbers

- Issuance grew >10× in under a decade. Early–mid 2005: $25–40 bn per quarter → peak ≈ **$100 bn per quarter** (late 2006–mid 2007) → first half of 2008: **< $5 bn** per quarter.
- By mid-2007: 37,000 AAA-rated structured finance issues in the US. About **60% of structured products were AAA** vs **<1% of corporate issues**.
- Moody's got **44%** of 2006 revenue from rating structured products. After the crash, rating-agency stock prices fell ~50%.
- Downgrades: 27 of 30 Merrill Lynch 2007 asset-backed CDO tranches fell from AAA to "junk". Moody's downgraded 31% of all ABS-CDO tranches in 2007 (14% of those rated AAA).

2. What is structured finance?

DEFINITION

Structured finance = **pool** many loans/bonds/mortgages together, then issue **prioritised claims** called **tranches** against the pool. Because of the priority order, many tranches are *far safer* than the average asset inside the pool.



Losses fill from the BOTTOM up (junior → mezzanine → senior).
Cash is paid from the TOP down (senior first).

Figure 12 – Pooling then tranching

EVERYDAY ANALOGY – A CHAMPAGNE TOWER / BUCKET STACK

Imagine three buckets stacked in a rain of losses: the bottom bucket (junior) fills first. Only when it overflows does the middle one start filling, and the top one (senior) gets wet only after both have overflowed. A senior holder only loses money if the pool suffers a *disaster*.

PASS-THROUGH VS TRANCHING

If the vehicle just gave everyone an equal fraction of the pool (**pass-through**), the rating would equal the pool's average rating — no improvement. Prioritisation (tranching) is what creates safer-than-average claims. The cushion beneath a tranche (junior claims) is called **overcollateralisation**; it sets how much loss the pool can take before the senior tranche is hurt.

3. The two-bond example (the heart of lecture 1C)

Take two identical bonds. Each pays 1 if it survives and 0 if it defaults; default probability p_D = **10%**, defaults independent. Pool them (total value 2) and issue two tranches of 1 each:

- **Junior** tranche: takes the first 1 of loss → it defaults if *either* bond defaults.
- **Senior** tranche: defaults only if *both* bonds default.

Outcome (independent)	Probability	Junior pays?	Senior pays?
Neither defaults	0.9 × 0.9 = 81%	1	1
Exactly one defaults	2 × 0.1 × 0.9 = 18%	0	1
Both default	0.1 × 0.1 = 1%	0	0

Junior default probability = 18% + 1% = **19%** (riskier than a single bond's 10%). **Senior default probability** = **1%** (ten times safer than a single bond!). Risk didn't vanish — it was *concentrated* in the junior tranche. Junior therefore sells cheap with a high promised return; senior sells dearer with a low promised return.

More bonds → even safer senior

Add a third bond (p_D = 10%, independent). Three tranches: first defaults if **≥1** bond defaults, second if **≥2**, senior only if **all 3**.

Tranche	Defaults if...	Probability
Junior	≥ 1 default	$1 - 0.9^3 = 27.1\%$
Middle	≥ 2 defaults	$3(0.1^2)(0.9) + 0.1^3 = 2.8\%$
Senior	3 defaults	$0.1^3 = 0.1\%$

Two-thirds of the capital (middle + senior) is now **safer than each underlying bond (10%)**. "Central insight": **the bigger the pool, the larger the share of tranches rated better than the pool's average**.

CDO-squared (CDO²)

Take the *junior* tranches (19% default) from two different two-bond pools, pool *those*, and tranche again. The new senior tranche defaults only if both juniors default: 0.19 × 0.19 ≈ **3.6%**. A **CDO** is "collateralised debt obligation"; a CDO of CDO-tranches is **CDO²**. You've manufactured a "safe-looking" asset out of the riskiest slices.

4. The catch #1: correlation

> **DEFAULT CORRELATION**
> How strongly defaults tend to happen *together*. Independent = 0; perfectly together = 1.

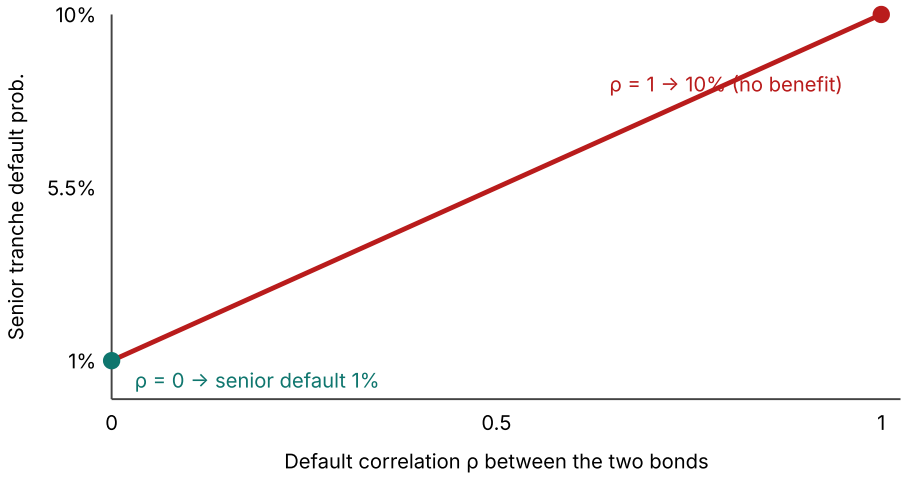


Figure 13 – Higher correlation → senior tranche loses its safety (two-bond example; line derived with the extra formula below)

EXTRA FORMULA (TWO BONDS, SAME P)

P(both default) = $p^2 + \rho \cdot p(1-p)$. For p = 10%: = 0.01 + ρ × 0.09. So ρ = 0 → 1%; ρ = 0.2 → 2.8%; ρ = 0.5 → 5.5%; ρ = 1 → 10%.

When defaults are perfectly correlated, both bonds live or die together and the senior tranche inherits the bond's 10% risk — **no credit enhancement at all**.

CATCH #2: RISK TYPE CHANGES

Securitisation swaps largely **diversifiable** risks for highly **systematic** ones. Senior tranches are safe in ordinary times, but in a deep economy-wide downturn (when defaults correlate) they fail. So they have far less chance of surviving a severe downturn than a normal corporate bond with the same rating — and investors should have demanded a **larger risk premium**.

5. Credit rating agencies

Moody's, Standard & Poor's and Fitch rate creditworthiness on the scale AAA, AA, A, BBB, BB... For over a century they rated **single-name** corporate debt (one company's bonds). Pre-crisis they expanded into **structured finance**. Issuers wanted their new products rated on the same scale as bonds so that investors with rating-based rules could buy them — creating an *illusion of comparability* with simple bonds.

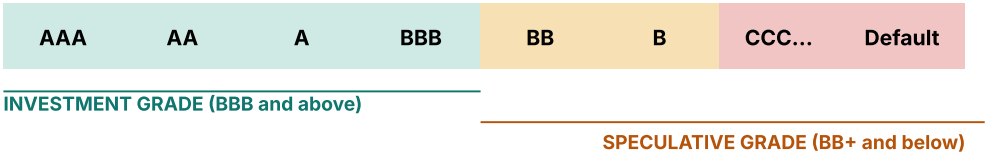


Figure 14 – Rating scale (simplified; real scale has + / – notches such as BBB–, BB+)

WHAT A RATING MEASURES

Either (a) the **probability of default**, or (b) the **expected economic loss** = probability of default × severity of loss given default.

$$\text{Expected loss} = \text{P(default)} \times \text{Loss given default}$$

Example (extra): P(default) 5%, bondholders recover 50% on default → loss given default = 50% → expected loss = 5% × 50% = **2.5%**.

6. Why rating structured products is so hard (Lecture 1D)

- For single-name bonds, each security is rated on its own — agencies didn't need to model whether defaults move together.
- For tranches they must model the **entire joint distribution** of defaults in the pool. Tranche riskiness is very **sensitive to default correlation**.
- CDO structures **magnify small errors** in default probability, correlation or model assumptions. After repeated rounds (CDO²) even errors too small to change an underlying bond's rating can swing the tranche rating massively. A "modest imprecision" can turn an AAA tranche into one that defaults with reasonable likelihood.

THE LECTURE'S SIMULATION

Setup: 40 CDO pools, each 100 bonds; each bond: 5-yr default prob 5%, recovery 50% → rating BB+ (just below investment grade). Pairwise default correlation 0.20 within a pool, 0 across pools. Tranches: **junior** absorbs losses up to 6%; **mezzanine** from 6% to 12%; **senior** above 12%. A CDO² pools the 40 mezzanine tranches and tranches again.

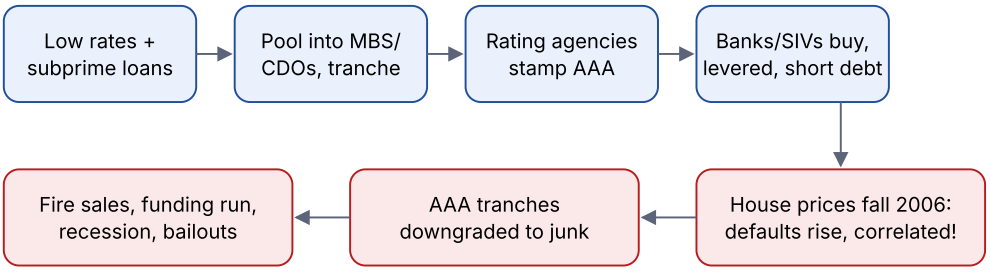
Result: original CDO: senior = **AAA**, mezzanine = **BBB−** (lowest investment grade). The CDO² built from mezzanine tranches: its mezzanine *and* senior tranches also reach **AAA** — from a pool of BBB− slices!

But: if the true correlation is higher than assumed, those tranche values fall sharply — and the CDO² is hit far harder than the original CDO.

WHY SUBPRIME MADE IT WORSE

Many CDOs held subprime mortgage-backed securities that were themselves tranches of earlier securitisations — effectively CDO²s. Subprime lending was recent, so there was **little historical data** on defaults; errors in default probabilities, correlations and recovery rates were likely and were magnified by re-securitisation → devastating losses.

7. The whole story in one picture



Boom → engineered “safety” → fragility → crash

Figure 15 – Summary of Parts D and E

PART F

Cheat Sheet

Concept	One-line summary
Value of asset	Depends on size of expected cash flows and uncertainty of forecasts.
Bond interest rate	i = (Face − P)/P ; P = Face/(1+i). Price ↓ when rates ↑.
Risky bond	Higher default risk → higher required rate → lower price.
Stock	Ownership claim on profits; dividends + retained earnings.
Systematic vs idiosyncratic	Market-wide (can't diversify, priced) vs firm-specific (diversifiable, irrelevant).
Required rate of return	Higher with more systematic risk → lower share value.
IPO / secondary market	First sale of shares to public / later trading between investors.
Limit order, bid, ask	Price-quantity commitment; bid = buy, ask = sell; book sorted best-first.
Bubble	Sustained, large gap between price and fundamental value. Fed by resale value, easy trading, easy borrowing, momentum.
EMH / Fama	Prices reflect all public information instantly; bubbles are mostly hindsight.
Shiller	Bubbles identifiable via e.g. P/E; “irrational exuberance”.
Leverage	Assets/equity; a 1/L fall in assets wipes out equity.
SIV	Off-balance-sheet vehicle funded by short-term debt that held securities; let banks dodge leverage rules.
Securitisation	Pooling loans into tradable securities (e.g. MBS).
Tranche / senior / junior	Prioritised slices of a pool; junior absorbs losses first.
CDO / CDO²	Debt securities backed by a pool of loans/bonds / backed by tranches of other CDOs.
Default correlation	Tendency of defaults to cluster; high correlation destroys tranche safety.
Investment grade	BBB and above. Speculative: BB+ and below.
Expected loss	P(default) × loss given default.
Policy tools 2008	Deposit insurance, Fed liquidity, TARP, zero rates, unconventional policy, fiscal stimulus.

Glossary (A–Z)

AAA	Highest credit rating (lowest default risk).
Aggregate demand	Total spending in the economy.
Fire sale	Forced selling at very low prices.
Illiquid	Hard to sell quickly without a big price cut.
Mortgage-backed security (MBS)	Security whose payments come from a pool of mortgages.
Overcollateralisation	Cushion of junior claims protecting senior tranches.
Recovery rate	Share of face value recovered after default.
Special purpose vehicle (SPV)	Separate entity holding the asset pool, isolating risk from the originator.
Subprime	Borrowers with weaker credit.
Wholesale funding	Short-term borrowing from other banks/investors (not insured depositors).
Zero lower bound	Limit that policy interest rates can't be cut meaningfully below zero.

PART F · PRACTICE

Practice Problems

Try each one on paper first. Full solutions start on the next page. E = easy, M = medium, T = think/explain.

Section 1 – Bonds, stocks, risk

Q1 E A one-year bond pays Rs. 10,000 in a year and currently costs Rs. 9,500. What interest rate does it yield?

Q2 E The market interest rate is 12%. What is the fair price of a one-year bond that pays Rs. 5,000 in a year?

Q3 M A one-year bond pays Rs. 10,000. Market rates rise from 8% to 10%. Compute the bond's price before and after, and say whether holders gain or lose.

Q4 T A government bond and a corporate bond promise identical payments. Which sells for less, and why? Which gives the higher return if no default occurs?

Q5 E A firm earns Rs. 20 crore with 2 crore shares and pays out 30% as dividends. You own 500 shares. (a) Earnings per share? (b) Your dividend? (c) Your share of retained earnings?

Q6 E Classify as systematic (S) or idiosyncratic (I): (a) RBI raises interest rates; (b) a pharma firm's new drug is approved; (c) a country-wide recession; (d) a fire at one firm's factory; (e) new tariffs on all imported goods.

Section 2 – Markets & bubbles

Q7 M Use the order book in Figure 6 (asks: 100 @ 101.00, 200 @ 101.10, 300 @ 101.50; bids: 150 @ 100.80, 200 @ 100.50). (a) You place a limit buy for 250 shares at 101.10. What trades, at what prices, and what is your total cost? (b) You place a limit buy for 100 shares at 100.90. What happens? (c) You place a limit sell for 200 shares at 100.50. What trades and how much do you receive?

Q8 M An index rises from 1,000 to 4,000 in 4 years, then falls to 800. (a) Average annual growth during the rise? (b) Percentage fall from peak to bottom?

Q9 T In 3–4 lines, explain why Fama says “bubbles are 20-20 hindsight” and how Shiller disagrees. Name the three features of markets that make bubbles more likely.

Section 3 – The 2008 crisis

Q10 M A bank has assets of Rs. 100 and equity of Rs. 4. (a) Leverage? (b) By what % must assets fall to wipe out equity? (c) What is left of equity (as % of the original) if assets fall 3%?

Q11 T A bank holds mortgages from all over the country, not just one city. Why wasn't that enough to protect it when US house prices fell nationwide? Mention correlation.

Q12 T List the three ingredients that made the financial system fragile in 2008, and explain how a SIV involved each.

Section 4 – Structured finance & ratings

Q13 M Two independent bonds each default with probability 20%. They are pooled and split into a junior tranche (defaults if either bond defaults) and a senior tranche (defaults only if both do). Find both tranche default probabilities.

Q14 M Add a third identical bond (20%, independent), giving junior (≥1 default), middle (≥2) and senior (3 defaults) tranches. Find all three default probabilities.

Q15 M Using the two-bond pool of Q13 but with the two bonds' defaults **perfectly correlated**, what is the senior tranche's default probability? Now take two of the *junior* tranches from independent Q13 pools and build a CDO² senior tranche that defaults only if both juniors default. What is its default probability?

Q16 M Two bonds, each $p = 10\%$, default correlation $\rho = 0.2$. Using $P(\text{both}) = p^2 + \rho \cdot p(1-p)$, find the senior tranche's default probability. How does it compare with $\rho = 0$ and $\rho = 1$?

Q17 E A bond has a 4% chance of default; if it defaults, investors recover 60% of face value. What is the expected loss?

Q18 E Investment grade or speculative grade? (a) AAA (b) BBB− (c) BB+ (d) A (e) B

Q19 T Why are senior tranches of a CDO "safe in normal times but dangerous in a crisis"? Use the words *systematic risk* and *correlation*.

Q20 T Why did rating CDO tranches require far more from the agencies than rating a single corporate bond? Give two reasons, including what goes wrong in a CDO².

Worked Solutions

A1. i = (10,000 − 9,500)/9,500 = 500/9,500 = **5.26%**.

A2. P = 5,000/1.12 = **Rs. 4,464.29**.

A3. At 8%: 10,000/1.08 = **9,259.26**. At 10%: 10,000/1.10 = **9,090.91**. Price falls by about 168; current holders **lose** (anyone buying afterwards gets the higher 10% yield).

A4. The **corporate** bond sells for less (investors demand a higher return for default risk, which means a lower price for the same promised payments). If it does not default, its buyer earns the higher return; but it might default, so that return is not guaranteed.

A5. (a) 20 crore / 2 crore = **Rs. 10 per share**. (b) Your profit share = 500 × 10 = Rs. 5,000; dividend = 30% × 5,000 = **Rs. 1,500**. (c) Retained = 5,000 − 1,500 = **Rs. 3,500**.

A6. (a) S (b) I (c) S (d) I (e) S.

A7. (a) Best asks: 100 @ 101.00 = 10,100; remaining 150 @ 101.10 = 15,165. Total cost = **Rs. 25,265** (average ≈ 101.06). Fully filled; 50 shares remain at 101.10 in the book. (b) 100.90 is below the best ask (101.00) so no trade: the order **enters the book** as the new best bid (above 100.80). (c) Sell at 100.50 or more matches bids: 150 @ 100.80 = 15,120, then 50 @ 100.50 = 5,025. Total received = **Rs. 20,145**.

A8. (a) Factor 4 over 4 years: $4^{1/4} \approx 1.414$ → **≈ 41.4% per year**. (b) 1 − 800/4,000 = **80% fall**.

A9. Fama (EMH): prices already incorporate all available information, so we can't know a price is "too high" before it falls; people who call bubbles are right about half the time, and we only celebrate the lucky ones afterwards. Shiller: prices can be driven by expectations of further rises ("irrational exuberance"); public stats like P/E can flag bubbles, and betting against them pays off in the long run. Three features: **resale value, ease of trading, ease of borrowing**.

A10. (a) 100/4 = **25×**. (b) 4/100 = **4%** (=1/25). (c) 3% fall → loss of 3 → equity 1 → **25%** of original equity remains (75% lost).

A11. Geographic diversification protects against *local* shocks (idiosyncratic risk). A nationwide fall is a **systematic** shock: mortgage defaults across regions become **highly correlated**, so the pool still suffers together. Securitisation also made originators care less about borrower quality.

A12. (1) **High leverage** – SIVs/banks held many assets on little capital (and SIVs let banks evade leverage rules). (2) **Illiquid assets** – SIVs held complex securities that could only be sold at fire-sale prices. (3) **Liquid liabilities** – SIVs were funded entirely by short-term wholesale debt, which investors could (and did) refuse to roll over; this debt wasn't covered by deposit insurance.

A13. Junior = $1 - 0.8^2 = 1 - 0.64$ = **36%**. Senior = 0.2^2 = **4%**.

A14. Junior (≥1) = $1 - 0.8^3 = 1 - 0.512$ = **48.8%**. Middle (≥2) = $3(0.2^2)(0.8) + 0.2^3 = 0.096 + 0.008$ = **10.4%**. Senior (3) = 0.2^3 = **0.8%**. (Middle and senior are both safer than a single 20% bond.)

A15. Perfect correlation: both bonds default together → senior default = **20%** (same as a single bond: no enhancement). CDO²: junior tranches default with 36%; independent pools → both default = 0.36^2 = **12.96% ≈ 13%**.

A16. P = 0.01 + 0.2 × 0.09 = 0.01 + 0.018 = **2.8%**. At ρ = 0 it's 1%; at ρ = 1 it's 10%. Even modest correlation nearly **triples** the senior default risk versus the independent case.

A17. Loss given default = 40%; expected loss = 4% × 40% = **1.6%**.

A18. (a) Investment (b) Investment (lowest notch) (c) Speculative (d) Investment (e) Speculative.

A19. In normal times defaults are few and scattered, so junior tranches absorb them and seniors are untouched. In a severe downturn, defaults become **correlated** (everyone is hit by the same economy-wide, **systematic** shock), the junior cushion is exhausted, and senior tranches start to lose — just when investors can least afford it. Diversification can't remove this risk.

A20. (1) Single-name bonds are rated one at a time; tranches need the **joint distribution** of defaults (probabilities *and* correlation), which is hard to estimate, especially for new subprime lending with little data. (2) Tranche ratings are very **sensitive** to small estimation errors, which structures magnify — in a CDO^2 errors compound over several rounds, so mistakes too small to change an underlying bond's rating can turn an AAA tranche into one that likely defaults.

> **HOW TO STUDY THIS**
>
> 1) Redo the bond formulas and the tranche probabilities from memory. 2) Redraw Figures 6, 9, 12 and 13 by hand. 3) For every crisis term, be able to say which of the three fragilities (leverage, illiquid assets, liquid liabilities) it belongs to. 4) Explain Fama vs Shiller in one minute.